

LAB 1: ADVANCED MEDICAL IMAGE PRE-PROCESSING METHODS

The goal of this lab practice is **to test** some of the advanced image **pre-processing algorithms** seen in class, to evaluate the results obtained with real clinical images, and to look for other types of image filtering methods available in the **state-of-the-art**.

To check the progress in each of the sections, **you can use examples of medical images of your choice (either from those available in Aula Virtual or obtained from the Internet)**. Some useful links for image search:

[GitHub - sfikas/medical-imaging-datasets: A list of Medical imaging datasets.](https://github.com/sfikas/medical-imaging-datasets)

<https://www.cancerimagingarchive.net/>

<https://nihcc.app.box.com/v/ChestXray-NIHCC>

<https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia>

[Open Access Medical Images - Finding Medical Images - Subject Guides at NYU Health Sciences Library](#)

Two different grayscale images must be selected, as they will be used throughout all sections of the assignment.

NOISE ADDITION

Since different image filtering techniques will be implemented, **it is necessary to evaluate the results obtained with noisy images**. To do this, **implement a simple algorithm** that adds some of the most important noise types: **Gaussian and impulsive**, with **3 different levels of intensity** (low, medium, high).

STANDARD SMOOTHING METHODS

In this first section, the **two images corrupted with the three levels of added Gaussian and impulse noise** must be filtered. To do so, a **mean filter**, a **Gaussian filter**, and a **median filter** will be used.

For each case, the **filter parameters that provide the best performance must be determined. Justify** their selection and **analyze** the results obtained.

NON-LOCAL MEANS FILTERING

Non-Local Means (NLM) method consists of filtering the image by weighted average of the different pixels in the image based on their similarity to the original pixel.

You should evaluate this filtering algorithm with the 2 selected images, with both noise type and the 3 levels of intensity. You must look for the parameters that work best for each scenario, justify your choice and **analyze** the results obtained. Finally, you should compare the filtering performance of this algorithm with the **standard smoothing methods** previously used.

If you are going to use Python, the algorithm is already implemented in the **denoise_nl_means** function of the **skimage.restoration** package. If instead MATLAB is used, the **imnlmfilt** function from the **Image Processing Toolbox** may be employed.

ANISOTROPIC FILTERING: PERONA AND MALIK

The goal of **anisotropic filtering** is to perform adaptive smoothing of the image based on its gradient. In this way, what we want to achieve is smaller smoothing in edge areas, and greater smoothing in homogeneous areas.

As well as in the first exercise, you must evaluate this filtering algorithm with same 2 images with both types of noise at three intensity levels. You should **look for the parameters that work best** with each of the image types and **justify** your choice. Also compare the results with **standard smoothing methods**.

You will have to look for an **implementation of this algorithm on the Internet** or **implement our own version**.

OTHER TYPES OF ADVANCED FILTERING

In this last exercise for this lab practice you must do some research of **other types of advanced filtering methods** that are available in the literature. You will have to perform an **in-depth explanation** of a method you like, to look for the **code implementation** (or to implement it if the code is not available), **to evaluate the results** with the same images as in the previous exercise, and **to compare its performance with the NLM, anisotropic filtering, and standard smoothing methods**.

SUBMISSION AND GRADING

For this lab practice it is requested to submit a **.zip file** with the following items:

- A **single PDF** containing the **report** of the completed assignment, with a **maximum length of 10 pages**. The methods used must be justified and the results obtained must be discussed, as indicated in the exercise statements.
- The **source code developed and used** to complete each exercise. You can use either Python or MATLAB.
- **The images used** throughout the assignments.

Since the assignment will be completed in pairs, the **report must clearly indicate the contribution of each team member, without prejudice to any additional checks that the professors may carry out** to ensure that all members have acquired the competencies expected from this practical assignment. In addition, the **use of AI tools** must be described if they have been used (i.e., in which sections and for what purposes). AI can be a useful tool for **assisting with code implementation**, but it **MUST NOT replace the students' programming work or judgment, NOR SHOULD IT BE USED for the interpretation or discussion of the results**.

Please remember that this is one of the last courses you need to pass before becoming a Biomedical Engineering graduate. Therefore, you should approach this assignment as if you were working in a hospital or a company and had to present it as part of your professional work. Consequently, the **final grade** will be based



on the quality of both, the **implemented or utilized code** (including appropriate explanatory comments) and the **report (including the explanation of the methods, their parameters and how they affect image filtering, as well as the analysis and discussion of the obtained results)**.

The **.zip file to be submitted** must be named as follows: **P1_MIA_BED_26_27_GroupX.zip**, where *GroupX* is the name of the group (for example, **P1_MIA_BED_26_27_G1_2.zip** would correspond to the submission of group **G1_2**).

The **submission deadline** is **October 22, 2026, at 23:55** for students carrying out the laboratory session on **October 8, 2026** (Groups **G1_x**), and **October 23, 2026, at 23:55** for students carrying out the laboratory session on **October 9, 2026** (Groups **G2_x**).